# Rajiv Gandhi University of Knowledge Technologies- RK Valley - AP

**Academic Year 2024 – 25: Semester 2 - Mid Semester Test-2**

**Computer Science & Engineering (CSE) - CS2202- OOPS Through JAVA**

**R22 Batch& Regular Mode of Examinations**

**Year:ENGG-1** | **Date of Examination:06/03/2025**

**Time:60 Minutes** R220714 | **Max Marks:15**

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**3X5 Marks= 15**

**The Question Paper consists of 6 questions. Students need to answer 3 questionsfrom each set**
**Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions**

1. Usage of BufferedReader and BufferedWriter. Write a java progra to display all odd numbered files of a text file.

**OR**

2. Write a java program Filter Out Special Characters from a String and the result Tokens are write into file. Given a string with various punctuation marks and special characters, use StringTokenizer to remove them and only print the alphanumeric tokens.
**Input:** "Hello! How's @everyone? Welcome... #Java"

3. What is multiple Inheritance? How to achieve it explain with Sample code.

**OR**

4. Explain about Polymorphism and Design a Payment base class with an abstract method makePayment().Create subclasses CreditCardPayment, DebitCardPayment, and UPIPayment that implement makePayment() differently.Demonstrate method overriding and runtime polymorphism.

5. What is user defined exception Handling? Write a Java program that simulates an online payment system. Handle exceptions for InsufficientBalanceException, InvalidCardException.

**OR**

6. A. What is exception handling and explain about exception handling techniques.[3M]

B. Develop a java program illustrates the application of multiple catch statements [2M]

Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2024 – 25: Semester II - Mid Semester Test-2

Computer Science Engineering (CSE)- 23MA1202 (Discrete Mathematics)(DM)

R22 Batch & Regular Mode of Examinations

Year: ENGG-1 | Date of Examination: 07-03-2025

Time: 60 Minutes | Max Marks: 15

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3X5 Marks= 15

The Question Paper consists of 6 questions. Students need to answer 3 questions from each set

Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

1) Out of 250 candidates who failed in an examination, it was revealed that 128 failed in mathematics, 87 in physics and 134 in aggregate. 31 failed in mathematics and in physics, 54 failed in the aggregate and in mathematics, 30 failed in the aggregate and in physics. Find how many candidates failed

(i) in all the three subjects (ii) in mathematics but not in physics (iii) in the aggregate but not in the mathematics (iv) in physics but not in the aggregate or in mathematics (v) in the aggregate or in mathematics, but not in physics.

**(OR)**

2) Below is the list of relations among people. For each of the following relations, state whether the relation is reflexive, symmetric, antisymmetric, and transitive

a) xRy stands for x is a child of y b) xRy stands for x is a spouse of y

c) xRy stands for x is a wife of y

d) xRy stands for x is a superior of y e) xRy stands for x and y have the same parents.

3) Let A = {1,2,3} and B = {a,b,c,d}. Let R and S be the relations from A to B with Boolean

Matrices $M_R=\begin{bmatrix}1 & 01 & 0\\ 0 & 10 & 0\\ 1 & 00 & 1\end{bmatrix}$ and $M_S=\begin{bmatrix}0 & 10 & 0\\ 1 & 00 & 1\\ 0 & 11 & 0\end{bmatrix}$

a) Find Boolean matrices for $R^{-1}$ and $S^{-1}$ [ 1 M ]

b) Find Boolean matrices for $(R\cap S)\bullet R^{-1}$ and $(R\bullet R^{-1})\cap(S\bullet R^{-1})$ [ 2 M ]

c) Find Boolean matrices for $S\bullet(R^{-1}\cup S^{-1})$ and $(S\bullet R^{-1})\cup(S\bullet S^{-1})$ [ 2 M ]

**(OR)**

4) If $f:R \to R$ and $g:R \to R$ defined by $f(x)=x^3-4x, g(x)=\frac{1}{(x^2+1)}$ , $h(x)=x^4$

Find the following composition functions.

a) $(f \circ g \circ h)(x)$ b) $(h \circ g \circ f)(x)$ c) $(g \circ g)(x)$ d) $(g \circ h)(x)$ e) $(f \circ g)(x)$

5) Draw the Hasse – diagram to illustrate the following partial ordering

a) The set of all subsets of {1,2,3,4} having at least two numbers partially ordered by $\subseteq$ [3M]

b) The set of all subsets of {1,2,3,4} having at most two numbe partially ordered by $\supseteq$ [2M]

**(OR)**

6) Use mathematical induction method to show that for all positive integers n, $3.5^{2n+1}+2^{3n+1}$ is divisible by 17

**Rajiv Gandhi University of Knowledge Technologies-RK Valley-AP**
**Academic Year 2024 – 25: Semester IIMidSemester Test-II**
**Computer Science & Engineering- Data Structures**
**R22Batch Regular Mode**

**Year: ENGG-I** **Date of Examination: 5-03-2025**
**Time: 60 minutes** R220714 **MaxMarks:15**

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3X5Marks=15

**The Question Paper consists of 6 questions with internal choice. Answer any 3 questions from each set. Each Question Carries 5 Marks**

**Note: Usage of Scientific calculator may be permitted**

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1) *What is a DeQueue? Explain its operation with Example?*

**OR**

2) Let Q denote a queue containing sixteen numbers and S be an empty stack. Head(Q) returns the element at the head of the queue Q without removing it from Q. Similarly Top(S) returns the element at the top of S without removing it from S. Consider the algorithm given below.

```
while Q is not Empty do
    if S is Empty OR Top(S) ≤ Head(Q) then
        x := Dequeue(Q);
        Push(S,x);
    else
        x := Pop(S);
        Enqueue(Q,x);
    end
end
```

What are the maximum possible number of iterations of the while loop in the algorithm?

3) Sort the following list of the elements by using Quick sort
9,6,7,1,8,5,4,2,3

**OR**

4) Sort the following list of the elements by using insertion sort
35,19,66,14,8,10,57,100

5) a)what is the result evaluating the postfix expression (2M)

10 5 + 60 6 / * 8 -

b) Explain binary search with example code (3M)

**OR**

6) a) Consider a hash table of size 11 that uses open addressing with linear probing. Let h(k) = k mod 11 be the hash function used. A sequence of records with keys 43 36 92 87 11 4 71 13 14 is inserted into an initially empty hash table, the bins of which are indexed from zero to ten. What is the index of the bin into which the last record is inserted?(2M)

b) Consider a hash table with 9 slots. The hash function is (k) k mod 9. The collisions are resolved by chaining. The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. What are the maximum, minimum, and average chain lengths in the hash table?(3M)